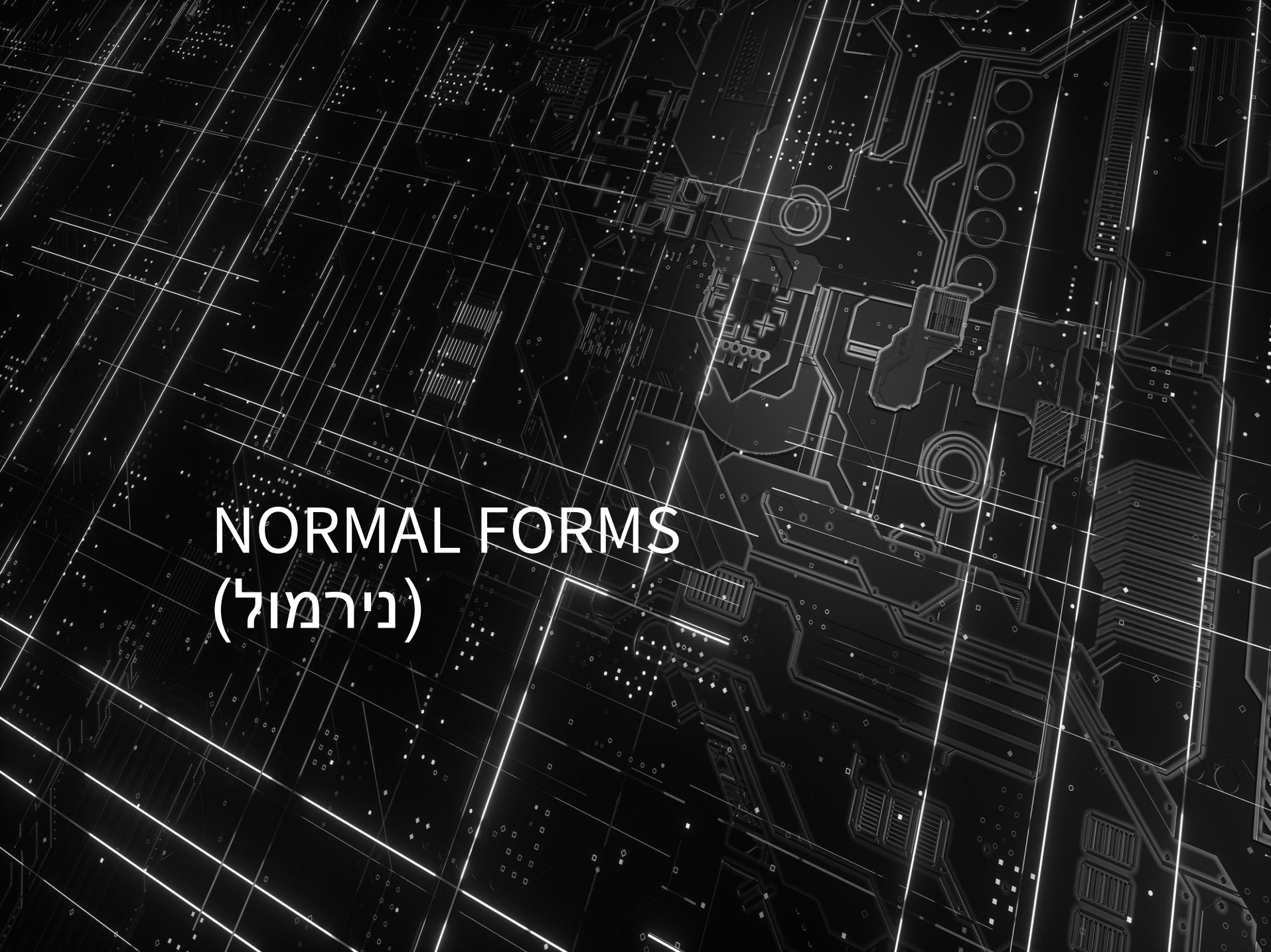




# NORMAL FORMS

(נירמול)

## MANY SCHEMAS – WHAT SHOULD I DO?

- יש מספר סכמות שיכולות לתאר "עולם" מסויים.
- הסכימה האידאלית צריכה לקיים:
- הצגת כל המידע הנדרש וקיום כל הקשרים.
- לא קיים מידע מיותר.
- ניתן לשנות מידע בצורה קלה.
- באמצעות תהליך הנירמול ניתן להגיע לסכימה האידאלית.

ישנן 6 רמות נירמול, אנחנו נעבור על ארבע מהן:

- First normal form (1NF)
- Second normal form (2NF)
- Third normal form (3NF)
- Boyce-Codd normal form (BCNF)

# FIRST NORMAL FORM (1NF)

- זיהוי קבוצות שחוזרות על עצמן
- העברתן לטבלה חדשה
- הגדרת מפתח לטבלה החדשה

# PROCESS 1NF

**Student\_Report** (StudentNo, StudentName, Major, CourseNo, CourseName, InstructorNo, InstructorName, Grade)

| StudentNo | StudentName | Major     | CourseNo | CourseName | InstructorNo | InstructorName | Grade |
|-----------|-------------|-----------|----------|------------|--------------|----------------|-------|
| 32170066  | Roni Gil    | Computers | 7032     | OS         | 23           | Moran Barak    | 80    |
| 32170066  | Roni Gil    | Computers | 7033     | Statistics | 48           | Moshe Tamar    | 76    |
| 35170063  | Shai Gover  | Business  | 7033     | Statistics | 48           | Moshe Tamar    | 76    |
| 35170063  | Shai Gover  | Business  | 7048     | English    | 51           | Silvi Mor      | 94    |

- In the **Student Report** table, the repeating group is the **course information**.

A student can take many courses.

- Remove the repeating group. In this case, it's the course information for each student.
- Identify the PK for your new table.

- The PK must uniquely identify the attribute value (StudentNo and CourseNo).
- After removing all the attributes related to the course and student, you are left with the student course table (**StudentCourse**).
- The Student table (**Student**) is now in first normal form with the repeating group removed.

## **Student** (StudentNo, StudentName, Major)

| <u>StudentNo</u> | StudentName | Major     |
|------------------|-------------|-----------|
| 32170066         | Roni Gil    | Computers |
| 35170063         | Shai Gover  | Business  |

## **StudentCourse** (StudentNo, CourseNo, CourseName, InstructorNo, InstructorName, Grade)

| <u>StudentNo</u> | <u>CourseNo</u> | CourseName | InstructorNo | InstructorName | Grade |
|------------------|-----------------|------------|--------------|----------------|-------|
| 32170066         | 7032            | OS         | 23           | Moran Barak    | 80    |
| 32170066         | 7033            | Statistics | 48           | Moshe Tamar    | 76    |
| 35170063         | 7033            | Statistics | 48           | Moshe Tamar    | 76    |
| 35170063         | 7048            | English    | 51           | Silvi Mor      | 94    |



## Problems that might be after 1NF:

- To add a new course, we need a student.
- When course information needs to be updated, we may have inconsistencies.
- To delete a *student*, we might also delete critical information about a course.

# SECOND NORMAL FORM (2NF)

- זיהוי עמודות שלא מבוססות על PK
- העברתן לטבלה חדשה
- הגדרת מפתח לטבלה החדשה

# PROCESS 2NF

Before:

| <u>StudentNo</u> | StudentName | Major     |
|------------------|-------------|-----------|
| 32170066         | Roni Gil    | Computers |
| 35170063         | Shai Gover  | Business  |

| <u>StudentNo</u> | <u>CourseNo</u> | CourseName | InstructorNo | InstructorName | Grade |
|------------------|-----------------|------------|--------------|----------------|-------|
| 32170066         | 7032            | OS         | 23           | Moran Barak    | 80    |
| 32170066         | 7033            | Statistics | 48           | Moshe Tamar    | 76    |
| 35170063         | 7033            | Statistics | 48           | Moshe Tamar    | 76    |
| 35170063         | 7048            | English    | 51           | Silvi Mor      | 94    |

- The **Student** table is already in 2NF because it has a single-column PK.
- In **Student Course** table, we see that not all the attributes are fully dependent on the PK; specifically, all course information.  
The only attribute that is fully dependent is grade.
- Identify the new table that contains the course information.
- Identify the PK for the new table.

## **Student** (StudentNo, StudentName, Major)

| <u>StudentNo</u> | StudentName | Major     |
|------------------|-------------|-----------|
| 32170066         | Roni Gil    | Computers |
| 35170063         | Shai Gover  | Business  |

## **CourseGrade** (StudentNo, CourseNo, Grade)

| <u>StudentNo</u> | <u>CourseNo</u> | Grade |
|------------------|-----------------|-------|
| 32170066         | 7032            | 80    |
| 32170066         | 7033            | 76    |
| 35170063         | 7033            | 76    |
| 35170063         | 7048            | 94    |

# CourseInstructor (CourseNo, CourseName, InstructorNo, InstructorName)

| <u>CourseNo</u> | CourseName | InstructorNo | InstructorName |
|-----------------|------------|--------------|----------------|
| 7032            | OS         | 23           | Moran Barak    |
| 7033            | Statistics | 48           | Moshe Tamar    |
| 7048            | English    | 51           | Silvi Mor      |

## **Problems that might be after 2NF:**

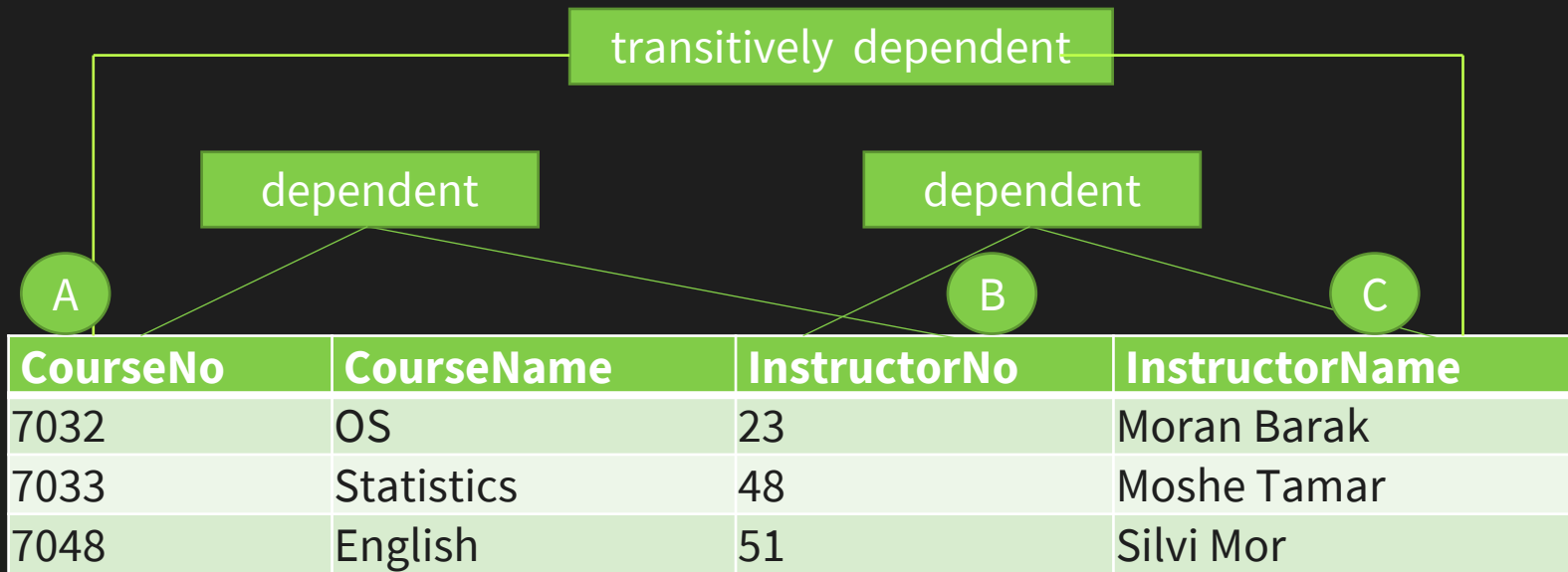
- When adding a new instructor, we need a course.
- Updating course information could lead to inconsistencies for instructor information.
- Deleting a course may also delete instructor information.

# THIRD NORMAL FORM (3NF)

- זיהוי עמודות טרנזיטביות ל PK
- העברתן לטבלה חדשה
- הגדרת מפתח לטבלה החדשה



# TRANSITIVE RELATIONSHIP



במידה ובטבלה את עמודה A שהיא מפתח ראשי ואת עמודות C ו B שהן לא מפתחות.  
כאשר יש תלות של C בעמודה B ועמודה B תלויה בעמודה A  
במקרה זה עמודה C ו A הן טרנזיטיביות

# THIRD NORMAL FORM (3NF)

a table is considered in third normal if the table/entity is already in the second normal form and the columns of the table/entity are **non-transitively** dependent on the primary key.

# PROCESS 3NF

Before:

| <u>StudentNo</u> | StudentName | Major     |
|------------------|-------------|-----------|
| 32170066         | Roni Gil    | Computers |
| 35170063         | Shai Gover  | Business  |

| <u>StudentNo</u> | <u>CourseNo</u> | Grade |
|------------------|-----------------|-------|
| 32170066         | 7032            | 80    |
| 32170066         | 7033            | 76    |
| 35170063         | 7033            | 76    |
| 35170063         | 7048            | 94    |

| <u>CourseNo</u> | CourseName | InstructorNo | InstructorName |
|-----------------|------------|--------------|----------------|
| 7032            | OS         | 23           | Moran Barak    |
| 7033            | Statistics | 48           | Moshe Tamar    |
| 7048            | English    | 51           | Silvi Mor      |

- Eliminate all dependent attributes in transitive relationship(s) from each of the tables that have a transitive relationship.\*
- Create new table(s) with removed dependency.
- Check new table(s) as well as table(s) modified to make sure that each table has a determinant and that no table contains inappropriate dependencies.

## **Student** (StudentNo, StudentName, Major)

| <u>StudentNo</u> | StudentName | Major     |
|------------------|-------------|-----------|
| 32170066         | Roni Gil    | Computers |
| 35170063         | Shai Gover  | Business  |

## **CourseGrade** (StudentNo, CourseNo, Grade)

| <u>StudentNo</u> | <u>CourseNo</u> | Grade |
|------------------|-----------------|-------|
| 32170066         | 7032            | 80    |
| 32170066         | 7033            | 76    |
| 35170063         | 7033            | 76    |
| 35170063         | 7048            | 94    |

**Course** (CourseNo, CourseName, InstructorNo)

| CourseNo | CourseName | InstructorNo |
|----------|------------|--------------|
| 7032     | OS         | 23           |
| 7033     | Statistics | 48           |
| 7048     | English    | 51           |

**Instructor** (InstructorNo, InstructorName)

| <u>InstructorNo</u> | InstructorName |
|---------------------|----------------|
| 23                  | Moran Barak    |
| 48                  | Moshe Tamar    |
| 51                  | Silvi Mor      |

# BOYCE CODD FORM (BCNF)

- זיהוי תלויות פונקציונליות בין עמודות
- העברתן לטבלה חדשה
- הגדרת מפתח לטבלה החדשה

# דוגמה:

| CustomerID | CustomerName | Loan ID | Amount |
|------------|--------------|---------|--------|
| 32170066   | Roni Gil     | 1       | 1000   |
| 35170063   | Shai Gover   | 2       | 5000   |
| 32170066   | Roni Gil     | 2       | 5000   |
| 35170063   | Shai Gover   | 1       | 1000   |



**Customer** (CustomerID, CustomerName)

| <u>CustomerID</u> | CustomerName |
|-------------------|--------------|
| 32170066          | Roni Gil     |
| 35170063          | Shai Gover   |

**Bor\_Loan** (CustomerID, LoanID, Amount)

| <u>CustomerID</u> | <u>Loan ID</u> | Amount |
|-------------------|----------------|--------|
| 32170066          | 1              | 1000   |
| 35170063          | 2              | 5000   |
| 32170066          | 2              | 5000   |
| 35170063          | 1              | 1000   |

**Problem:**

Updating the amount information could lead to inconsistencies for loan id.

**Customer** (CustomerID, CustomerName)

| <u>CustomerID</u> | CustomerName |
|-------------------|--------------|
| 32170066          | Roni Gil     |
| 35170063          | Shai Gover   |

**Bor\_Loan** (CustomerID, LoanID)

| <u>CustomerID</u> | <u>Loan ID</u> |
|-------------------|----------------|
| 32170066          | 1              |
| 35170063          | 2              |
| 32170066          | 2              |
| 35170063          | 1              |

**Loan** (LoanID, Amount)

| <u>Loadn ID</u> | Amount |
|-----------------|--------|
| 1               | 1000   |
| 2               | 5000   |

# תרגיל:

- נתונה הטבלה הנ"ל.
- יש לבנות טבלאות לפי חוקי הנירמול.

| elevator no | building no | building name | capacity | staff no | first name | last name | date examined |
|-------------|-------------|---------------|----------|----------|------------|-----------|---------------|
| 3           | 1           | Arthium       | 15       | 1        | Moshe      | Oren      | 01.05.2021    |
| 6           | 1           | Arthium       | 15       | 1        | Moshe      | Oren      | 02.05.2020    |
| 18          | 3           | Azrieli       | 20       | 4        | Amir       | Doron     | 02.04.2020    |
| 16          | 3           | Azrieli       | 12       | 4        | Amir       | Doron     | 02.04.2020    |